

REPORT

Basic Network Scanning Using Nmap

Name: Himanshi Trivedi

Internship: Oasis Infobyte (OIBSIP)

Task: Basic Network Scanning with Nmap

Tool Used: Nmap

1. Introduction

Network scanning is a fundamental activity in cybersecurity used to identify active devices, open ports, and running services on a system. Open ports can expose a system to potential security risks if they are not properly secured. This task demonstrates the use of Nmap to perform a basic network scan on a local machine.

2. Objective

The objectives of this task are:

- To perform a basic network scan using Nmap
 - To identify open ports and services on the local system
 - To understand the security significance of open ports
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3. Tool Description

Nmap (Network Mapper) is an open-source network scanning and security auditing tool. It is widely used to discover hosts, detect open ports, identify running services, and analyze network exposure.

4. Methodology

The following steps were followed to complete this task:

1. Installed Nmap on the system
2. Opened Command Prompt
3. Executed the command `nmap localhost`
4. Observed the open ports and services
5. Saved the scan results and captured screenshots
6. Analyzed the security implications of the findings

5. Scan Command Used

```
nmap localhost
```

6. Findings (Open Ports Identified)

Port	Service	Description
135	MS RPC	Windows Remote Procedure Call service
445	SMB	Windows file and printer sharing service
1521	Oracle	Oracle database listener
3306	MySQL	MySQL database service
5500	Hotline	Oracle Enterprise Manager
49165	Unknown	Dynamic Windows service port
61900	Unknown	Ephemeral system port

7. Analysis

The scan results indicate that several Windows system services and database services are running on the local machine. Ports 135 and 445 are commonly used by Windows for internal communication and file sharing. Ports 1521 and 3306 indicate active Oracle and MySQL database services. The high-numbered ports are dynamically assigned by the operating system for temporary communication.

8. Security Significance

- Open database ports can expose sensitive data if not secured
 - Windows services such as RPC and SMB are common attack targets
 - Dynamic ports increase the attack surface if left unmonitored
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9. Recommendations

To enhance system security, the following measures are recommended:

1. Disable unused services
 2. Restrict database access using firewall rules
 3. Use strong authentication mechanisms
 4. Regularly monitor open ports using Nmap
 5. Keep the system and applications updated
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10. Conclusion

This task demonstrated how Nmap can be used for basic network scanning to identify open ports and running services. Understanding open ports helps in recognizing potential security risks and applying appropriate protective measures.